

Examining the Impact of Video Reality-level to Support Transition from Screen Time to Screen-Free Time (TSSF)

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ABSTRACT

Transition from Screen time to Screen-Free time (TSSF) refers to stopping video-watching and transiting to other offline activities. Motivated from priming effect, we explore how video reality-level, which is the degree of using real elements in the video such as the backgrounds and characters, can affect children's TSSF. Children aged 3-10 were assigned to watch videos with three different reality-levels: low, mid, and high. A total of 83 parent-child dyads participated in a two-week between-subject experimental study. We analyzed four different aspects of TSSF: transition rate, problematic behavior, screen time dependence, and rule compliance. Our experiment showed that compared to children who watched low-reality videos, children who watched high-reality videos exhibited more successful TSSF in three aspects: transition rate, problematic behavior and screen time dependence.

CCS CONCEPTS

• **Human-centered computing; Human computer interaction (HCI); Empirical studies in HCI;**

KEYWORDS

video reality-level, screen time, transition to screen-free time, problematic behavior, screen time dependence, rule compliance

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1 INTRODUCTION

Children's screen time from digital media is increasing with the prevalence of Internet platforms such as YouTube [4]. Video-watching became a dominant activity of screen time for children, as users can now easily watch videos at any time and in any place. However, the negative impact of exposure to excessive amounts of digital media such as reduced feelings of social acceptance [26],

problematic behaviors, and cognitive developmental delay [19] has been observed. Given that media usage habits in childhood can persist even through adulthood [13], existing research examined methods for promoting healthy media consumption behavior at a young age [12]. One important step in promoting such a healthy media consumption behavior in young children is to support a smooth "Transition from Screen time to Screen-Free time (TSSF)", which is transitioning from video-watching to offline activities.

Existing methods have mainly focused on supporting TSSF by applying interventions pre- and post-video-watching activities. For example, one commonly used pre-intervention strategy includes setting screen time rules *before* video-watching [16]. Although an effective strategy, setting video-watching rules and educating children until positive video-watching habits are formed can be difficult for some parents. An example of a common post-intervention strategy is suggesting alternative activities *after* video-watching. Previous research demonstrated that suggesting children's favorite offline activities, such as reading books, helped smooth TSSF [15, 28]. However, if the presentation of alternative activities are not offered properly, children may receive negative impressions and think that they are forced to perform other activities [30]. Therefore, given the potential difficulties of existing pre-/ post-interventions, we explore an intervention strategy for supporting successful TSSF *during* video-watching activities.

1.1 Priming effect strategies for TSSF: During video-watching

Although applying intervention during video-watching to support TSSF has not been explored much, strategies to help other types of children's behavior, such as promoting healthy eating behavior [22], during video-watching have previously been reported. In such an approach, videos that contain the desired behavioral contents are presented to children [8]. The underlying theory for the success of such an approach is the use of the priming effect, which is a phenomenon whereby an exposure to a certain stimulus influences a response to a subsequent stimulus, without any awareness of the connection [29]. In the context of video-watching, engaging in video-watching serves as a primer for children's desired behavior, thereby resulting in the desired behaviors that are expected of children following the video-watching. In this paper, we expand on such previous studies to examine whether the video priming method can also be applied to support successful TSSF.

In order to apply video priming to children's video-watching context, we suggest using video reality-level to influence children's desired behavior. We define video reality-level to be the degree of

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Figure 1: Example screens from three video reality-levels¹

using “real” elements of characters or backgrounds in a given video. The reality of the video is divided into three levels: low-, mid-, and high-reality. In a video, the two elements that can have realistic features are 1) fictional characters/objects, and 2) background setting. Specifically, a low-reality video features fictional characters or personalized objects that appear in an unreal setting, such as animation videos that are created using graphics only. A mid-reality video refers to a video in which real and unreal elements exist together. A mid-reality video may also show the activity of a real person in an unreal environmental setting, such as an animated background. Finally, a high-reality video features real people, animals, or objects in a real background. Figure 1 shows example screens of three video reality-levels.

In prior study, videos that use many unreal elements, i.e. low-reality videos, has been shown to be effective for creating engaging learning environments [1]. Therefore, we conjecture that watching videos with a low reality level would make it difficult for children to stop watching the video and to transition to reality, compared to when they watch videos with high-reality levels. Our research question is summarized as follows: How does video reality-level affect children’s TSSF?

1.2 Children’s behaviors in video-watching

In previous studies that observed children’s behaviors related to video-watching, researchers showed successful transitions through proposed intervening systems [27]. Here, a successful transition is how well a child transitions into a real world after promised video-watching time. For example, Shin and colleagues proposed an augmented reality (AR) character-embedded system which suggested alternative activity such as playing offline hide-and-seek when the set video-watching time was reached. As the system successfully provided a means to connect the real world and video content with alternative activity, a smooth transition to real world was achieved [28]. Within our research context, we expect that if the video content is closer to a realistic offline environment in which children can engage after video-watching, they could transition to the real world with more ease.

Hypothesis 1 (H1): The higher the video reality-level, the higher the transition rate.

Existing research consistently reports that video-watching activity is positively correlated with problematic behavior in children [25]. For example, a meta-study that analyzed the behavioral outcomes of children who watched television showed that increased screen time tends to increase children’s problematic behavior such as stereotypes, and materialism [14]. Given the evidence on how increase in screen time is associated with increase in problematic behavior, we expect that when children watch high-reality videos, they will be involved in less amount of screen time, which in turn will be associated with less problematic behavior.

Hypothesis 2 (H2): The higher the video reality-level, the less screen-time related problematic behavior.

Another aspect of children’s behavior that has been examined in connection to video-watching is screen time dependence, which is the children’s level of positive attitude such as preference, engagement, and affinity on video-watching compared to offline activities like playing with their siblings, drawing, or outdoor activities [11]. Previous studies showed that with an increase in the amount of screen time watched by children, positive attitudes such as preference, engagement and affinity for screen time increase as well. A novel aspect of our study is examining how a video reality-level can affect the screen time dependence. Since we expect that children who watch high-reality videos would engage in less amount of screen time, they would also exhibit less amount of screen time dependence.

Hypothesis 3 (H3): The higher the video reality-level, the lower the level of screen time dependence.

Rule compliance is also one type of a behavior that has been studied in children’s video-watching. Existing studies defined rule compliance as an ability of a child to follow rules on their own [11]. To measure rule compliance, researchers have examined levels of children’s self-regulation as well as compliance to parental rules [6]. For example, a longitudinal previous study on the relationship between the amount of video-watching time and self-regulation in children reported a negative association [5]. Additionally, the level of compliance with parental screen time rules are also reported to have a negative correlation with the amount of screen time [24]. Taken together, the evidence of a negative correlation between screen time and rule compliance suggests that when children watch high-reality video that leads to decreased screen time, children’s level of rule compliance would increase.

Hypothesis 4 (H4): The higher the video reality-level, the higher the level of rule compliance for screen time.

¹Low level of video reality: <https://www.youtube.com/watch?v=skSbg1Gup8>
Mid level of video reality: <https://www.youtube.com/watch?v=29TR33moqUA>
High level of video reality: <https://www.youtube.com/watch?v=5if4cjO5nxx>

2 STUDY METHOD

We conducted a between-subject field study for two weeks to examine whether successful TSSF would appear differently depending on the video reality-level. The independent variable was the video reality-level, which was provided at three different levels: low-, mid-, and high-reality.

2.1 Study participants

A total of 83 parent-child dyads participated in our study: 27 dyads in the low-reality video group (13 boys and 14 girls), 29 dyads in the mid-reality video group (12 boys and 17 girls), and 27 dyads in the high-reality video group (14 boys and 13 girls). To recruit participants, we posted introduction of our study and recruitment announcement on online application that is used to socialize with neighbors. We also adopted snowball sampling method, in which the participants recommended other participants. A \$50 gift certificate was provided to each household as a reward.

We selected our study participants to be children aged 3-10 years who watched videos more than twice or one hour a week. To prevent differences in the video-watching behavior depending on the family environment of each participant, we targeted families of the middle class and above in the metropolitan area [23]. To determine the demographic similarity of three video reality-level groups, the parent participants provided their children's age ($m=6.78$, $SD=2.28$) and video-watching time ($m=74.7$ min, $SD=28.13$). No statistically significant differences were observed among the three groups.

2.2 Experimental procedure

Prior to the main experiment, we examined the validity of three video reality-levels. We conducted a study with five parents, who are different from the parents of the main experiment. During this validity study, we presented the definition of three video reality-levels and showed three example videos for each reality-level. Subsequently, the parents were asked to evaluate the reality-levels of 10 YouTube videos. Finally, the kappa values showed 0.899 calculated using the responses of the participants and golden set from researcher, which indicated a high level of agreement, as a value of 0.8 or higher means almost perfect agreement [20].

The overall process of the main experiment is presented in Table 1. The main experiment consisted of a pre-meeting, video-watching period, and a post-meeting. During the pre-meeting, we introduced the study to the parents and their children, and informed consent was obtained through approved experimental form (IRB No. 2012/003-001). Then, the parents conducted a pre-survey regarding problematic behavior, screen time dependence, and rule compliance. The parent-child participants were also asked to establish video-watching time rules together. Each dyad was then randomly assigned to one of the low-, mid-, or high-reality video groups. Then parents received instructions on the video reality-levels and tested with mini exercise of 10 sample videos so that the parents could distinguish the assigned video reality-level from the other two levels. In addition, to ensure compliance with video quality guidelines, we sent daily reminders to watch only assigned reality-level videos. We also verified that the participating children only watched videos corresponding to assigned reality-levels during follow-up interviews. Note that the participants could watch

any video from existing platforms such as Youtube or Netflix, as long as the selected videos corresponded to the assigned video reality-level. During the two-week video-watching period, all participants submitted a daily survey on TSSF. In the post-meeting, questionnaires on problematic behavior, screen time dependence, and rule compliance were completed by the parents. A follow-up interviews were also conducted to gain additional insights into the children's TSSF behaviors with each parent participant.

2.3 Measurements of TSSF

The four video-watching behavior variables that are used to measure TSSF are 1) transition rate, 2) problematic behavior, 3) screen time dependence, and 4) rule compliance. First, the transition rate is a measurement of how well a child transitions into the real world after the promised video-watching time ends and calculated as the ratio of "the number of days with successful transition" over "the number of days when a child watched videos." A given day is considered to be successful transition day if a parent assigned a rating of 4 or 5 on a 5-point Likert scale question. Because a parent is asked to assign one rating per day, the parent is asked to consider a child's video-watching behavior throughout all the multiple video-watching sessions that could occur during a given day.

We measured three additional variables as indicators of children's video-watching behaviors; problematic behavior, screen time dependence, and rule compliance. A problematic behavior is defined as the level of children's misbehavior that was caused by video-watching. Examples of problematic behaviors included throwing tantrums when a parent asked the child to stop watching videos. A screen time dependence means the children's level of preference for video-watching compared to other activities such as playing with their siblings, drawing, reading books or outdoor activities. Lastly, rule compliance is a measure of a child's ability to follow rules related to video on their own. To measure these three types of behavior, we used measurements that were originally developed and verified by the National Information Society Agency in South Korea [23], and was modified by Shin and colleagues [28]. Each of the three video-watching behavior variables was measured with three questions that were answered on a 5-point Likert scale (1: strongly disagree, 2: disagree, 3: neutral, 4: agree, 5: strongly agree). Sample questions for each behavior variables are as follows: "I often argue with my child regarding video-watching usage (problematic behavior)", "My child always wants to watch videos (screen dependence)", "My child adheres to the usage rule without my enforcement (rule compliance)". The total scores were calculated as an average of the responses of the three questions, resulting in a maximum of 5 points.

In addition, in order to understand the qualitative aspects of the four variables, a total of five questions were asked to the parents of the participants during post-interview: two about transition rate (e.g., "How is your child's transition to offline activity like?"), one about problematic behaviors ("Did your child throw tantrums when she/he was asked to end the video-watching?"), one about screen time dependence ("How much does your child enjoy video-watching compared to other activities?"), and one about rule compliance ("Did your child follow the video-watching time rule

Table 1: Experimental procedure

Process	Task	Measurements
Pre-meeting	Pre-survey of video-watching behavior Video-watching time rule setting Assign video reality-level group	Problematic behavior, Screen time dependence, Rule compliance
Video-watching period	Daily survey on TSSF	Transition rate
Post-meeting	Post-survey of video-watching behavior Follow-up interview	Problematic behavior, Screen time dependence, Rule compliance

Table 2: ANOVA results for the four measurements of TSSF

Measurements		LR (n=27)	MR (n=29)	HR (n=27)	ANOVA	Tukey post-hoc
H1: Transition rate	Mean (SD)	0.59 (0.17)	0.69 (0.20)	0.88 (0.13)	F=23.38***	LR<MR<HR
H2: Problematic behavior	Pre-post diff (SD)	0.54 (0.56)	-0.04 (0.75)	-0.73 (0.63)	F=25.61***	LR>MR>HR
	Mean_post (SD)	2.73 (0.91)	3.02 (1.07)	2.86 (0.78)		
H3: Screen time dependence	Pre-post diff (SD)	0.27 (0.74)	0.00 (0.67)	-0.59 (0.62)	F=11.46***	LR=MR>HR
	Mean_post (SD)	2.38 (0.74)	2.56 (0.97)	2.62 (0.87)		
H4: Rule compliance	Pre-post diff (SD)	-0.19 (0.59)	-0.02 (0.59)	0.10 (0.69)	F=1.41	-
	Mean_post (SD)	3.90 (0.80)	3.86 (0.85)	3.90 (0.95)		

LR: low-reality, MR: mid-reality, HR: high-reality video group ***p < 0.001, **p < 0.01, *p < 0.05

during the study?”). Furthermore, the post-meeting included two questions on episodes about children’s behavioral changes (e.g., “Could you tell me if there are any differences in your child’s behavior compared to the start of the study?”) and two questions on comments regarding the video reality-level (e.g., “What are your thoughts on the relation between the video reality-level and the child’s video-watching behavior?”).

3 RESULT & DISCUSSION

We present and discuss the results of our experiment to examine whether a high video reality-level supports successful TSSF as Table 2. We report the results of the ANOVA for the four measurements of TSSF: transition rate (H1), problematic behavior (H2), screen time dependence (H3), and rule compliance (H4). An ANOVA test was used for statistical analysis of the differences among the four measurements across the video reality-level groups. In all tests, the differences between the experimental groups were considered as significant when the p-value of the statistical analysis was less than 0.05. In addition, qualitative analysis was performed using the data obtained through the follow-up interview from the post meeting. We adopted template analysis, which is used to understand the context in a wide range of subjects, and to analyze the experiences and perspectives of study participants [3].

3.1 Transition rate (H1)

Children in the high-reality video group exhibited a higher transition rate compared to those in the low- and mid-reality video groups (H1). The ANOVA test revealed a significant difference among the groups (F(2,80)=23.38, p<0.001). A post-hoc Tukey revealed that children in the high-reality video group had a significantly higher

transition rate than the low- (diff.=0.30, p<0.001) and the mid-reality video groups (diff.=0.19, p<0.001).

The high rate of the transition in the high-reality video group may be due to the videos suggesting alternative offline activities after watching videos. Because the activities in the videos that children watched are similar to ones in the real world, such activities can easily be transferred to the real world. In the follow-up meetings, parents said that the high-reality videos suggested alternative activities, such as origami and making flying a paper airplane. “When we reach promised watching time, my daughter willingly turned off the tablet and we started to discuss what she had watched while doing other activities like drawing or playing with toy blocks. Sometimes she wanted to draw a picture that she watched from the videos.” (p-11, high-reality)

However, low-reality videos did not seem to suggest alternative activities after video-watching, so a successful transition to offline may be difficult. Parents in the low-reality video group said that they had to make additional efforts to stop their children from watching videos, such as letting them know in advance when they would end video-watching or suggesting offline activities to enjoy. “To transit to screen free time, I (mother) had to prepare snacks that my daughter likes or suggest other activities such as board game or toy play. It was too difficult to stop watching if I (mother) did not offer such alternative activities.” (p-12, low-reality)

Previous studies also reported that suggesting an alternative activity was helpful for a successful transition, and that an intervention system that suggested reading books or playing seek-and-hide supported smooth transitions for children [9]. In our experiment, the high-reality videos might have served as an intervention system, acting as alternative activities suggestions as in the previous studies.

3.2 Problematic behavior (H2)

The difference in problematic behavior among the three groups was significant, as hypothesized ($F(2,80)=25.61$, $p<0.001$). Statistical analysis for the whole population showed that watching a higher video reality-level resulted in less problematic behavior compared to those who watched mid- and low-reality video groups. A post-hoc Tukey test showed that children in the high-reality video group had a significantly lower problematic behavior than the low- (diff.=1.27, $p<0.001$) and the mid-reality video groups (diff.=0.69, $p<0.001$). In addition, children in the mid-reality group showed lower problematic behavior than low-reality group (diff.=0.58, $p<0.001$).

The follow-up interview data presented the decrease in problematic behavior in the high-reality video groups. Parents reported that when their children watched high-reality videos, it was easier than before to transfer to other playing activities in the TSSF situation, so the problematic behaviors that had previously occurred were reduced. *“When there was no one to play with, he always wanted to watch videos. However, it seemed like there were fewer cases of stubbornness during last week. I think he participated in more physical activities like origami and Lego blocks because he usually watched videos about making something.”* (p-6, high-reality)

The increased problematic behavior in the low-reality video group compared to mid-reality group might have stemmed from the children being too immersed in the videos. Parent participants in the low-reality video group reported that their children were so focused on the video that it was difficult for them to get back into the real world. *“While watching a video, my son did not even notice if I (mother) talked to him or sometimes he completely forgot the promise to end the screen time. In addition, the video-watching activity also interfered with his daily lives, such as eating and going to bed.”* (p-9, low-reality)

3.3 Screen time dependence (H3)

The ANOVA analysis between the three reality-level showed a significant difference on screen time dependence ($F(2,80)=11.46$, $p<0.001$). Specifically, children in the high-reality video group had a significantly higher transition rate than the low- (diff.=0.86, $p<0.001$) and the mid-reality video groups (diff.=0.59, $p<0.001$). One possible explanation for the high screen time dependence of the low-reality video group is that children felt close to the characters. Existing study have reported that children regard virtual characters to be more friendly than real people [18]. In those studies, researchers also found that children showed more interest in the messages conveyed by the characters and wanted to keep engaging with the characters through video-watching rather than engage in alternative offline activities. In our experiment, many of the animated characters in low-reality videos may appear friendlier to children because their appearance and behavior are similar to those of children. Such children’s intimacy and preference towards the main character may have increased their engagement level in video-watching activity. We observed the children’s preferences and intimacy with the characters in the follow-up interviews. *“My son really likes the characters in the videos and wants not only watch them but also to buy toys related to the theses characters. Also, he frequently mimics the speech patterns and actions of cartoon characters.”* (p-24, low-reality)

3.4 Rule compliance (H4)

Unlike our hypothesis, our experiment result did not show significant difference in rule compliance according to the video reality-levels ($F(2,80)=1.41$, $p=0.25$) (H4). We observed a ceiling effect, which occurs when participants’ scores cluster toward the best possible score of the measure prior to the experiment [10]. From the pre-survey of video-watching behavior, the score of rule compliance from all participants was 3.89 (SD=0.86). These scores are quite high compared to the other measurements of problematic behavior ($m=2.88$) and screen time dependence ($m=2.52$). Therefore, we suspect that participants of our study already exhibited a high level of rule compliance prior to the study, so that no substantial improvement was observed from our study. In fact, the participants who had relatively low level of rule compliance before our study showed an increase in rule compliance scores. Thus, we anticipate that conducting a future study with participants who have a lower level of rule compliance may show a difference before and after the experiment.

4 CONCLUSION

Our study using video priming to aid TSSF presents two novel findings that have not been demonstrated in previous studies. First, our results extend existing studies on the use of video content for behavioral change, demonstrating that video priming is possible through using various video elements, such as the reality-level. As a result, we showed that in order to impact children’s video-watching behavior, changing the video content is not a requirement. In fact, using video reality-level has certain advantages over using video content as a factor that can impact children’s video-watching behavior. Second, our study expands the use of video priming. Video priming has been applied to behavioral research such as social skills [2], functional skills [17], occupational skills [7], exercise abilities, and emotional control in children. Studies using video priming theory for TSSF are rare, and our experimental study is one of the earliest attempts to show that video priming can be used for TSSF in children.

Although present study offers a promising potential to support TSSF in children without an external intervention system, our results exhibit certain limitations. First, the video story that was watched may have affected the results. In our experiment, the participants were randomly assigned to one of the low-, mid-, and high-reality video groups. While participants were instructed to view only the videos corresponding to their assigned video reality-level, there were no restrictions on the stories they could choose to watch. Second, the demographics of the participating children may have influenced the outcome. Although our study targeted children aged 3–10 years in the metropolitan area, we did not restrict factors such as parenting styles. Existing studies demonstrated that parenting attitudes or existence of siblings have an impact on the development of children’s emotions and behaviors [21]. Therefore, examining additional demographic factors could yield additional insights on TSSF, particularly on the rule compliance variable, in which we did not observe a difference.

Based on our results of the study, there are three points to consider when developing proper screen time habit for children. First, when developing system to support children’s TSSF, designers could

consider reality-level of video elements, such main characters, background setting, in order to provide connection with the real world. Second, we recommend that parents show high-reality videos to their child in their leisure time. Since the video reality-level has a relatively clear concept, parents can easily apply at home without a separate intervention applications or parental monitoring systems. Last but not least, researchers could develop educational program that introduce the concept of video reality-level with sample videos so that parents or teachers could utilize them in their educational environment.

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